LungCT AI | CLINICAL NOISE REPORT (Model 4 (NoiseCNN))

Date: 2026-09-03 08:53:02
Protocol: Model 4 (NoiseCNN)

Case / Scan ID:	LCT-85725	**Patient Case:**	ANONYMOUS (Clinical Study)
File Name:	test_ct_slice.png	**Analysis Model:**	NoiseCNN Classification & FFT Spectrum Analysis
Modality:	Computed Tomography (CT)	**Report Status:**	**COMPLETED (AI-Generated)**

CT Scan Imagery & Diagnostic Overlay

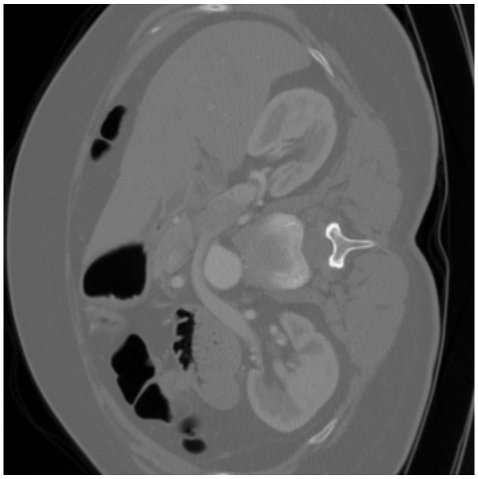

Figure 1: Original CT Scan
Input diagnostic image

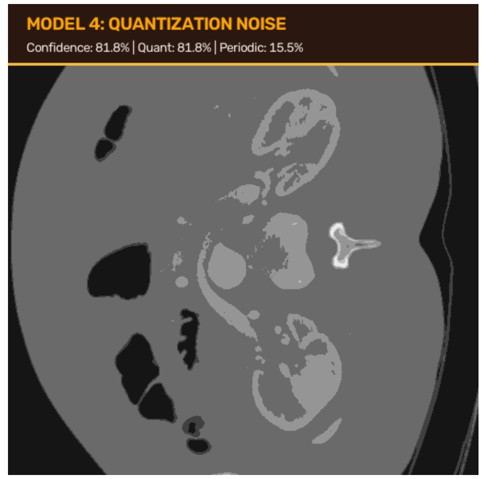

Figure 2: Diagnostic View & 2D Fourier Magnitude Spectrum

Quantitative Noise Classification & Metrics (Model 4 (NoiseCNN))

Noise Classification	Pixel Count / Prob.	Severity / Score	Clinical Classification
Quantization Noise (ADC Step)	Probability: 81.8%	81.77%	**SEVERE**
Periodic Noise (FFT Spike)	Probability: 15.5%	15.48%	**MILD**
Scan Status / Confidence	Predicted: Quantization Noise	81.8% Conf.	**QUANTIZATION NOISE**
TOTAL NOISE PROBABILITY (Model 4)	**Noise Likelihood**	**97.25%**	**CRITICAL**

Clinical Interpretation & Denoising Recommendations

Model Assessment (Model 4 (NoiseCNN)): Elevated noise artifacts detected (97.2%). Structural detail is noticeably compromised by noise patterns.
Radiologist Action: Strongly recommend executing deep neural restoration or wavelet-based denoising. If diagnostic ambiguities persist, evaluate sensor hardware.

Radiologist Review Block	AI System Validation Specs
______________________________ Reviewing Radiologist Signature	**Architecture:** NoiseCNN Deep Classifier (Vasanth.pth) **Spectrum:** 2D Fast Fourier Transform (FFT) **Classes:** Quantization, Periodic Noise

Disclaimer: This report was automatically generated by Model 4 (NoiseCNN) for multi-noise classification and severity assessment in educational & research environments. Findings should be cross-referenced by a qualified medical professional prior to clinical intervention.